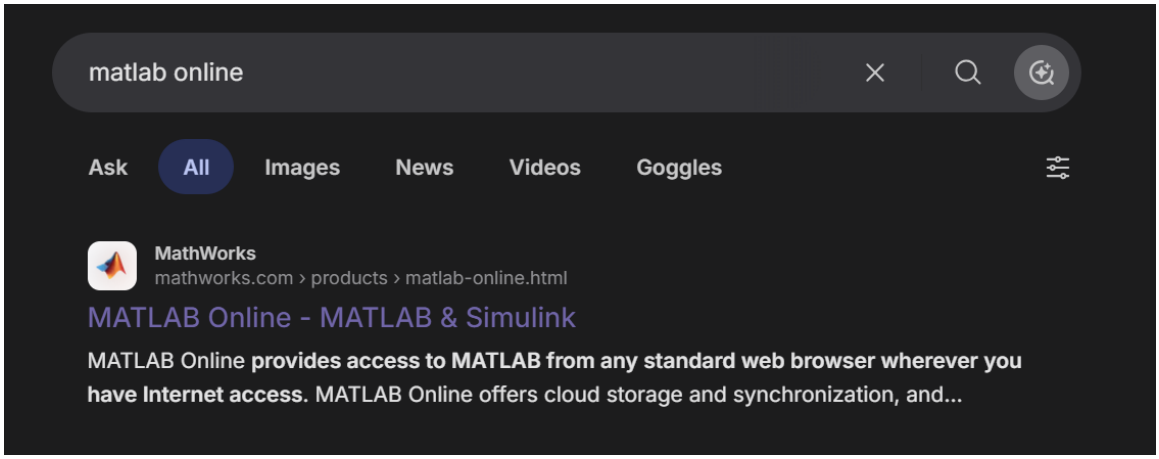
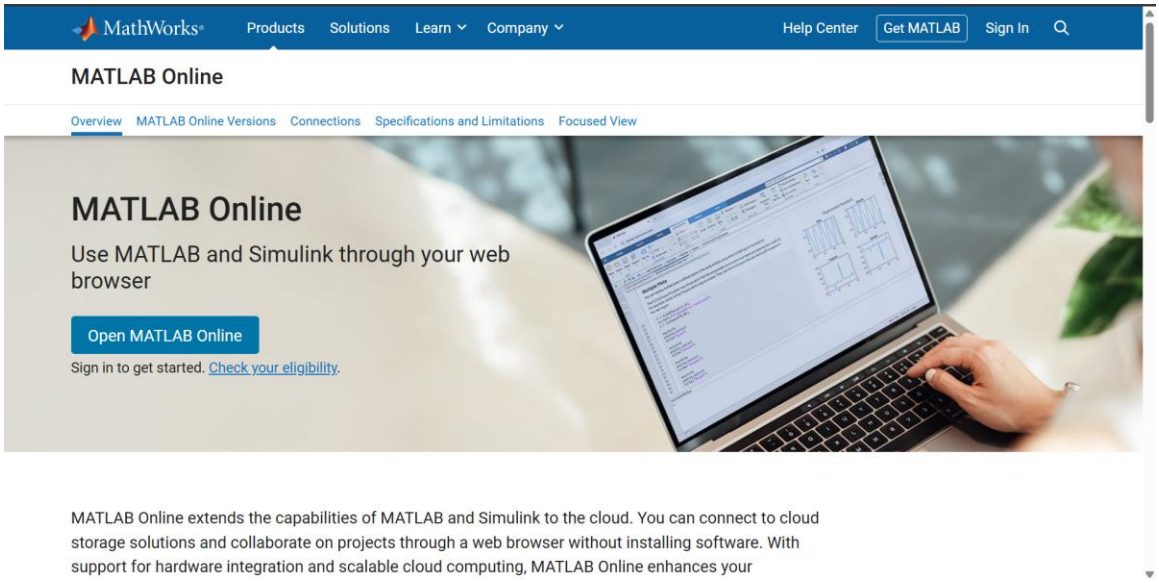
MATLAB Sign-In Guide

Step-by-Step Instructions: -

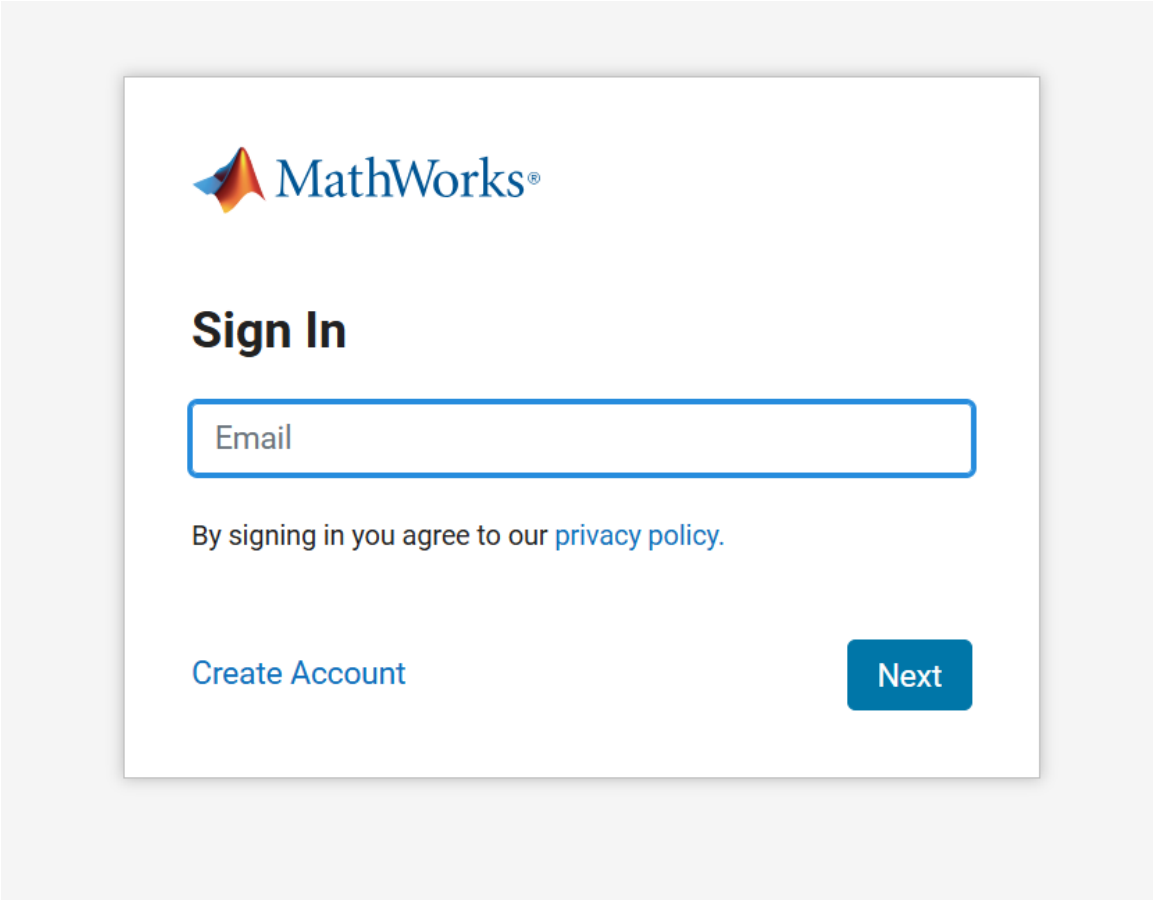
Step 1: Open MATLAB Portal: Open the MATLAB website or application as shown in the screenshot.



Step 2: Locate and click the 'Sign In' button (top-right corner).



Step 3: Enter Email: Enter your registered (personal) email ID.



Step 4: Enter Password

→ Enter or set your password correctly.

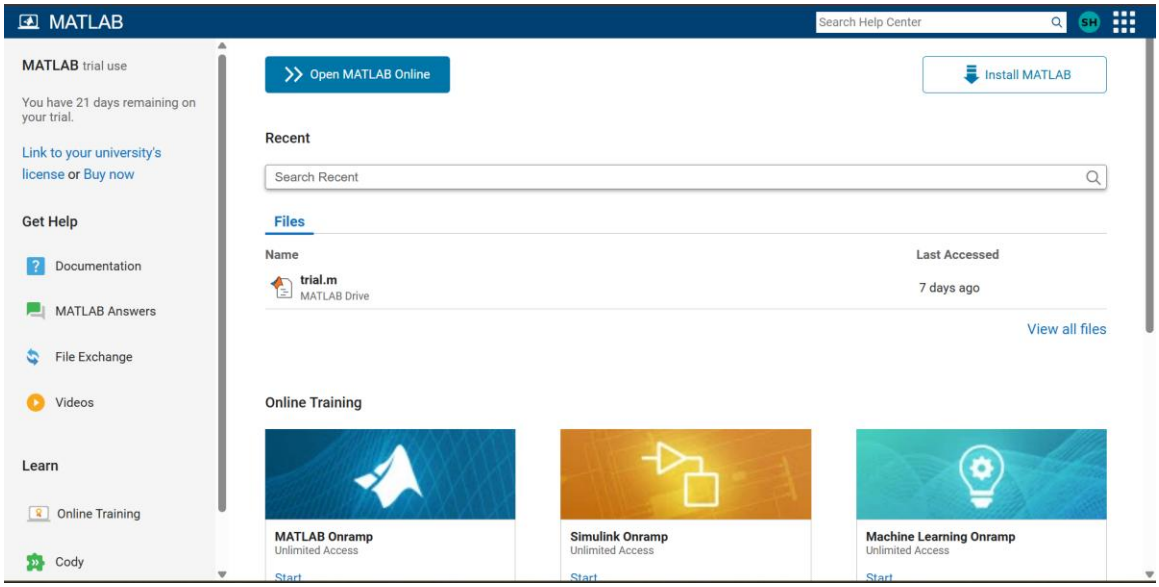
Step 5: Login

→ Click on 'Next' or 'Login'.

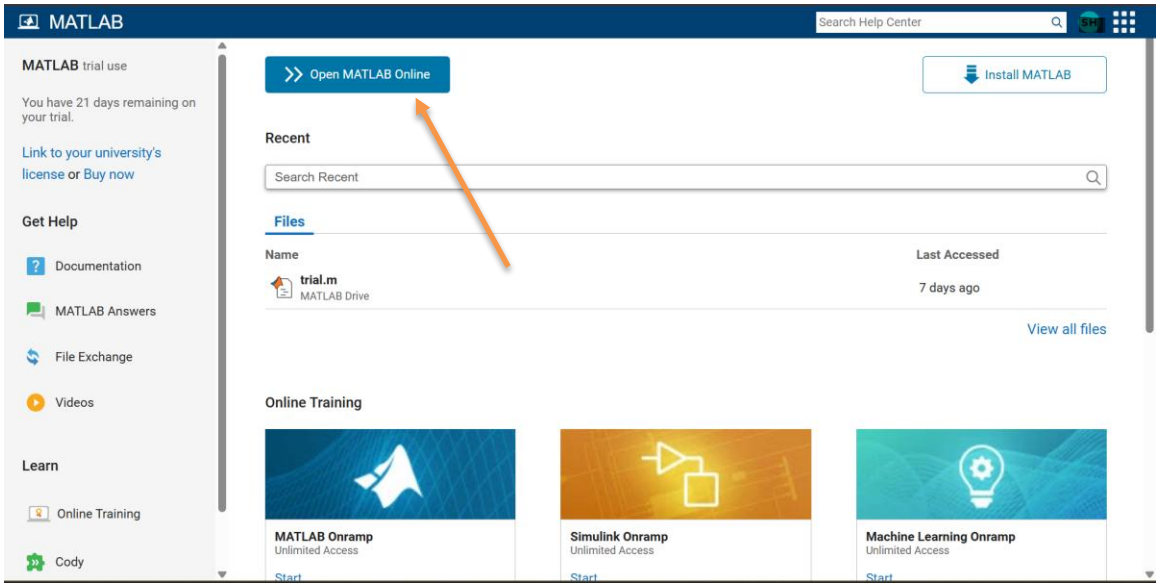
Step 6: Verification

→ Complete OTP or institutional verification if prompted.

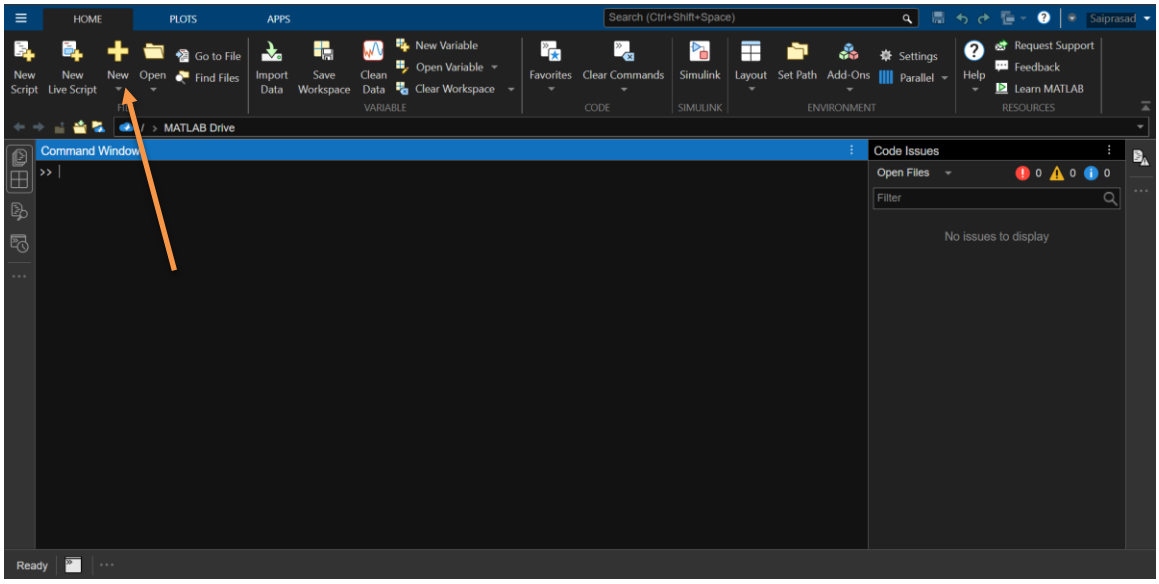
Step 7: Dashboard Access: After logging in, the MATLAB dashboard will be displayed.



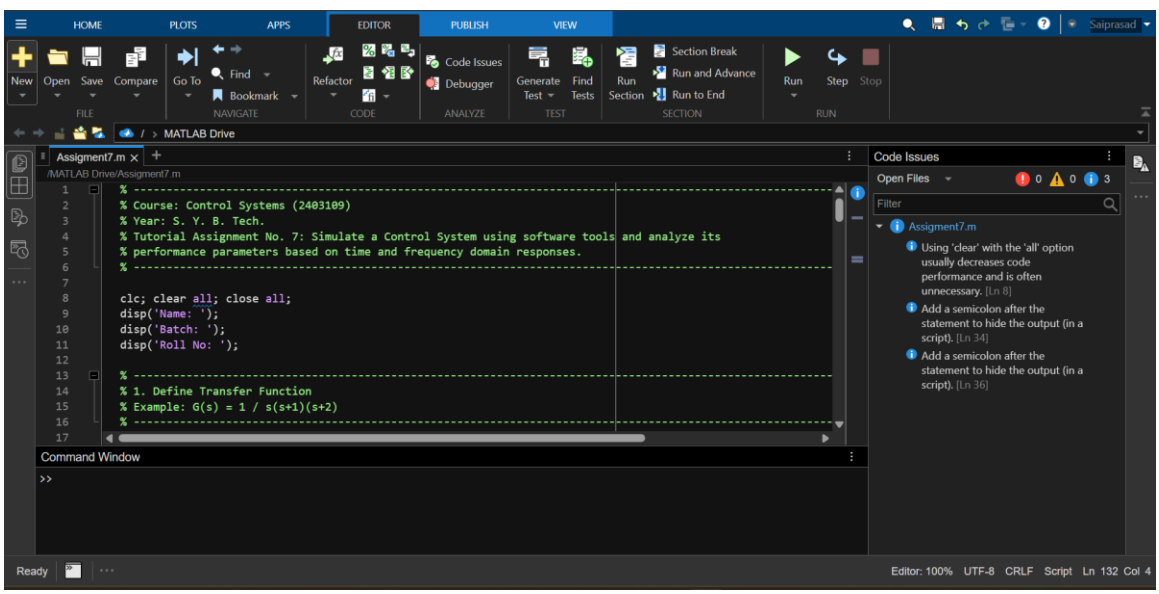
Step 8: Launch MATLAB: Click 'Open MATLAB' or 'Launch MATLAB Online'.



Step 9: Click on the 'New' option.



Step10: Paste the code from the shared .m file.



Step11: Enter the required inputs to obtain the output.

```
Command Window
Name:
Batch:
Roll No:
Enter number of denominator factors: 3
Enter factor 1 (e.g. [1 0] for s, [1 1] for s+1): [1,0]
Enter factor 2 (e.g. [1 0] for s, [1 1] for s+1): [1,1]
Enter factor 3 (e.g. [1 0] for s, [1 1] for s+1): [1,2]
Enter number of numerator factors: 1
Enter factor 1 (e.g. [1 0] for s, [1 1] for s+1): [1]
```

Output:

Open-loop Transfer Function G(s):
G =

```
      1
 -----------------
 s^3 + 3 s^2 + 2 s
```

Continuous-time transfer function.
Model Properties
Closed-loop Transfer Function T(s):

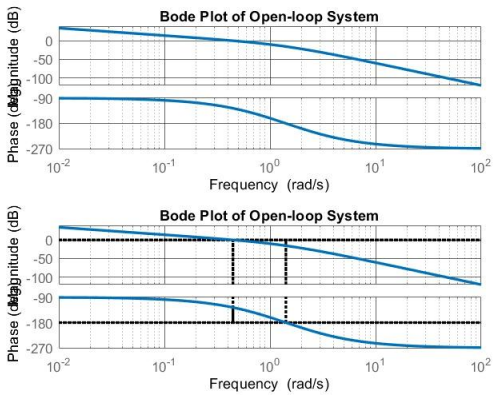
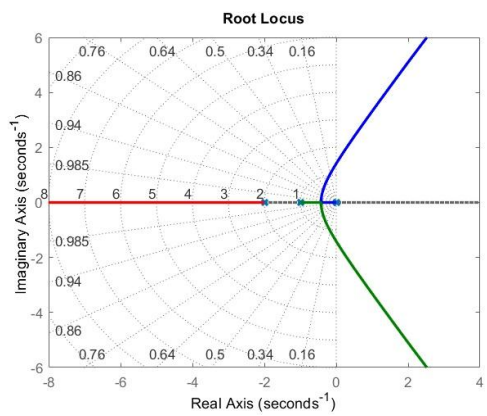
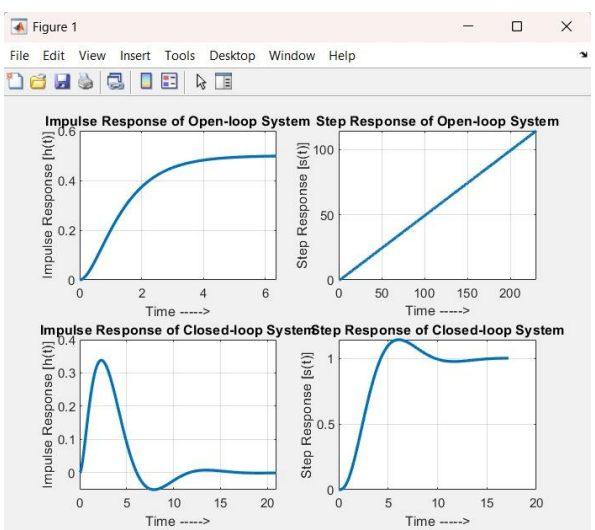
T =

```
        1
 ---------------------
 s^3 + 3 s^2 + 2 s + 1
```

Continuous-time transfer function.
Model Properties

Time Domain Specifications of Closed-loop System (Performance Parameters):
RiseTime: 2.6792
TransientTime: 12.3572
SettlingTime: 12.3572
SettlingMin: 0.9140
SettlingMax: 1.1445
Overshoot: 14.4509
Undershoot: 0

Peak: 1.1445
PeakTime: 6.1377

Closed-loop Poles:
-2.3247 + 0.0000i
-0.3376 + 0.5623i
-0.3376 - 0.5623i

System is Stable.